JAI-HUA KEVIN YEN

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EDUCATION

The State University of New York at Buffalo *August 2024 - May 2028 (expected)*
Ph.D Candidate in Biostatistics

National Taiwan University *August 2017 - August 2019*
Master of Science in Agronomy (Biometry Divison)

National Tsing Hua University *August 2013 - June 2017*
Bachelor of Science in Quantitative Finance

SKILLS

- **Programming Languages:** Python, R, SQL, SAS, C++, MATLAB, Golang
- **Machine Learning Tools:** Pytorch, Tensorflow, PaddlePaddle
- **Tools & Platforms:** Linux, Docker, Git, LaTex, Airflow
- **Core Competencies:** Explainable Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Causal Inference, Longitudinal Data Analysis, Machine Learning Operations (MLOps), Edge Computing

INDUSTRY EXPERIENCE

QNAP Taipei, Taiwan
Senior AI Software Engineer *February 2022 - June 2024*

- Trained a ResNet50 image classification model with a 70% F1-Score across 250 image classes and optical character recognition (OCR) models using vision transformer architecture with 90% accuracy to enable efficient customer photo searches
- Leveraged CLIP (Contrastive Language-Image Pre-Training), a multimodal model, to integrate semantic search into the customer's local files search engine, empowering customers to search for photos and files using human-readable descriptions instead of conventional keywords
- Deployed models on edge devices, including customer's local systems with Tensor Processing Unit (TPU) and Neural Processing Unit (NPU) acceleration, improving data security and inference speed within constrained computing resources
- Mentored junior personnel on programming, project organization, and training OCR models

TutorABC Taipei, Taiwan
AI Algorithm Engineer *May 2020 - February 2022*

- Created an 81% accurate customer churn detection model using random forest to decrease churn rates by 5%
- Developed an XGBoost customer purchase detection model with a 78% accuracy rate; utilized explainable machine learning method (SHAP value) to boost purchase rates and formulate targeted sales strategies
- Extracted key customer feedback using computer vision (YOLO, OCR) and natural language processing (TFIDF, BERT) models, enhancing data analysis for additional product features and improved user experience
- Accelerated the extraction, transformation, and loading (ETL) processes by 50%; consolidated by implementing Airflow for machine learning operations (MLOps)
- Mentored a junior engineer in programming, project structuring, and MLOps

RESEARCH EXPERIENCE

National Taiwan University Taipei, Taiwan
Research Assistant *August 2017 - August 2019*

- Derived an adjusted Chao2 species richness estimator, reducing underestimation by 15% and mitigating a 10% overestimation in scenarios with very low corrected denominators while mitigating identity errors common in species surveys
- Conducted data cleaning on weed and plant cover species surveys conducted in Soft Bridge County, Taiwan, and Grand St. Bernard Pass, Switzerland
- Developed a user-friendly R-Shiny application to make the research data and findings accessible to non-statistical audiences

TEACHING EXPERIENCE

The State University of New York at Buffalo New York, USA
Teaching Assistant *August 2024 - Now*

National Taiwan University & Taipei Medical University Taipei, Taiwan
Teaching Assistant Conference Lecturer *Feburary 2019*

National Taiwan University Taipei, Taiwan
Teaching Assistant *September 2017 - January 2019*

PUBLICATIONS

Yen, J.-H., Chiu, C.-H. (2020). Richness estimation with species identity error. *Proceedings 62nd International Statistical Institute World Statistics Congress*, **Volume 6**, 401-408.

Yen, J.-H., Chiu, C.-H. (2019, August 18-23). *Richness estimation with species identity error* [Poster presentation]. The 62nd International Statistical Institute World Statistics Congress, Kuala Lumpur, Malaysia.

Chiu, C.-H., **Yen, J.-H.** (2018, July 22-27). *Richness estimation in the existence of species identity error* [Poster presentation]. The 61st Symposium of the International Association for Vegetation Science, Bozeman, Montana, USA.